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This paper addresses the problem of verifying safety and finite-time reachability for discrete-time nonlinear systems controlled by neural networks with ReLU activation functions. The paper first studies backward reachable sets and proposes computing interval outer enclosures by solving mixed-integer linear programs, where the nonlinear dynamics are overapproximated using polytopic sets (see Optimization Problem (6)).

The authors then consider the problem of underapproximating backward reachable sets starting from the computed interval outer enclosures, and propose several variants, namely SHARP, CRISP, and CLEAN.

Subsequently, the paper introduces a verification framework called FABRIC, which combines backward and forward reachability analysis. In this framework, the time horizon is divided into two segments: forward reachability analysis is performed over the first segment, starting from the initial set, while backward reachability analysis is conducted over the second segment, starting from the target set. Verification succeeds if the overapproximated forward reachable set at the intermediate time step is contained within the underapproximated backward reachable set at the same step, thereby guaranteeing finite-time reachability. Moreover, safety is certified if the overapproximated forward and backward reachable sets remain disjoint from the obstacle sets throughout the considered time horizon. The authors then illustrate their approach through several benchmark numerical examples, demonstrating the effectiveness of the proposed framework.

Overall, the problem under consideration is interesting and highly relevant, and the paper is generally clearly written, which is appreciated. However, the manuscript currently lacks sufficient mathematical rigor, and several parts of the exposition remain vague and lacking in technical detail, which weakens the presentation of the novelty and contributions of the work. In my opinion, the paper could be considered for acceptance after major revision. The following are my comments, several of which are significant.

- The authors fail to adequately acknowledge or discuss state-of-the-art backward and forward reachability methods for neural control systems. See, for example, the recent works of Xiangru Xu and coauthors:
  - Zhang, Y., & Xu, X. (2026). *Forward and Backward Reachability Analysis of Closed-loop Recurrent Neural Networks via Hybrid Zonotopes*. arXiv preprint arXiv:2603.11547.

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– Zhang, Y., Zhang, H., & Xu, X. (2024). *Reachability analysis of neural network control systems with tunable accuracy and efficiency.* IEEE Control Systems Letters, 8, 1697–1702.

– Zhang, Y., Zhang, H., & Xu, X. (2023). *Backward reachability analysis of neural feedback systems using hybrid zonotopes.* IEEE Control Systems Letters, 7, 2779–2784.

The authors should acknowledge such works and clearly explain, at least conceptually, the advantages of the proposed framework relative to existing state-of-the-art methods.

- Equation (1): the considered system has the form

$$x + \delta(f(x) + u(x) + w),$$

where x is the state, w is the disturbance, and δ is a time step. This formulation appears highly restrictive, as it effectively assumes full-dimensional matched control and disturbance channels. Moreover, the model seems to originate from an Euler discretization of a continuous-time system. Existing methods in the literature typically consider more general classes of nonlinear systems with neural controllers. I believe the system formulation should be generalized to improve the applicability and relevance of the proposed approach.

- Above Equation (4), the notation "$i \in [1, T]$" does not appear consistent with the notation previously introduced for discrete intervals.

- Equations (4a) and (5a): the properties described in (4a) and (5a) are not equivalent. Property (4a) implies that the number of steps required to reach the target may depend on the initial state in I, whereas (5a) implies that all states in I reach the target in the same number of steps. Thus, (4a) is strictly weaker than (5a). It is therefore confusing that the paper presents them as equivalent verification objectives.

- Section 3: "In contrast, our setting does not impose ..." In practice, input constraints are typically essential. It is therefore unclear why the absence of input bounds should be viewed as an advantage. The authors should clarify this point.

- Section 3: "We call solution to Equation (3) ..." In my opinion, Equations (2) and (3) provide definitions rather than equations to be solved. I suggest rephrasing accordingly.

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- Equation (7): it is unclear why overapproximating the two-step backward reachable set is particularly advantageous. Why not consider three-step or general n-step backward reachable sets? Additional discussion and justification are needed.

- Equation (8): the volume operator will generally lead to a nonlinear optimization problem, while the forward reachability operator itself is typically nonlinear with respect to the set $\mathcal{X}^i$. How do the authors guarantee that the resulting optimization problem is solved formally, in the sense that the obtained solution rigorously satisfies the constraints? Moreover, how is the issue of local minima addressed?

- Algorithm 2: the placement appears inappropriate. Please verify whether the current location is intended.

- Above Equation (9): the notation

$$\mathcal{X}^o = x \pm r$$

  is not a mathematically rigorous definition of a hyperrectangle. Please revise this notation.

- SHARP and CRISP approaches: these approaches rely on box underapproximations, which will likely result in significant conservatism. It is unclear how such methods can outperform existing state-of-the-art approaches based on more expressive set representations, such as hybrid zonotopes or constrained zonotopes. The authors should discuss this issue more carefully.

- Evaluation: I would highly appreciate if the authors selected one representative example and illustrated graphically the backward reachable sets, together with how safety and reachability are certified. In addition, I encourage the authors to compare, for a single step, the accuracy and conservatism of their backward reachable set approximation against a hybrid-zonotope-based method. I do not necessarily expect the proposed approach to outperform such methods, but I would like to better understand the tradeoff between computational simplicity and conservatism.

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